

Roy (Yuanxi) Cheng

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EDUCATION

University of California, Los Angeles (UCLA)

Sep 2023 – Mar 2027

Bachelor of Science in Statistics and Data Science, Minor in Data Science Engineering

GPA: 3.75

- **Awards & Programs:** Dean's Honor's List, Persolv AI Intensive, BCG Case Comp Semi-finalist
- **Certificates:** Data Science & SQL in Python, EDA in R, Data Manipulation with dplyr, Data Visualization with ggplot2
- **Relevant Coursework:** Data Analysis & Regression, Machine Learning, Neural Networks & Deep Learning, Data Mining, Data Structures & Algorithms, Probability & Statistical Inference, Databases, NLP, Monte Carlo, Optimization

PROJECTS AND RESEARCH

Biomedical Tissue Imaging | *Python, PyTorch, YOLOv8, Pandas, NumPy, Matplotlib, OpenCV*

- Published article evaluating CNN-based instance segmentation architectures with AP metrics, benchmarking cytoplasm boundary precision for biomedical imaging
- Converted 4k+ CytoNuke annotations to COCO format, standardizing training across 3 segmentation architectures
- Trained YOLOv8 instance segmentation models in PyTorch, improving cytoplasm AP50 by 5% over Cyto-RCNN baseline

COVID Pandemic Effects on Secondary Education | *Python, Pandas, Tableau, Excel, Matplotlib*

- Cleaned 100k+ student records across multi-level datasets with pandas, enabling accurate longitudinal analysis in Tableau
- Aggregated 4 years of test scores into time-series visualizations, revealing student performance impacts tied to pandemic
- Created interactive Tableau map linking geographic data to score changes across U.S. states for a public research website

Adrenaline and Cognitive Fatigue Effects on Athletic Performance | *R, ggplot2, tidyverse, dplyr*

- Designed 2³ factorial experiment with 96 participants across 8 conditions, blocking by age and gender to reduce variance
- Conducted three-way ANOVA and Tukey post hoc tests in R, revealing no significant interactions at $\alpha = 0.05$
- Validated ANOVA assumptions using log transforms and ggplot2 diagnostics, confirming normality and homoscedasticity

WORK AND LEADERSHIP EXPERIENCE

Capital One

McLean, Virginia

Product Management and Analysis Intern

Jun 2026 – Present

- Analyze corporate workforce data with SQL and Python, identifying effects and factors tied to successful org transitions
- Process 1.5k BA performance reviews through a semantic similarity pipeline, reducing 923 O*NET occupations to 200+ DWA candidates via FAISS and ranking to 31 skills by prevalence and LLM evidence score
- Build composite scoring model ranking DWAs by frequency and LLM-assigned evidence strength, delivering a recruiter-facing skills menu to reframe job qualifications from bottom-up review data

Kohl's

Los Angeles, California

Project Manager and Data Analyst

Jun 2025 – Mar 2026

- Analyzed 8.5M+ logistics records with SQL and Python, resolving data quality to build a reliable shipment-level dataset
- Conducted EDA in Vertex AI Workbench, engineering supply-side shipment features that sharpened transit time forecasts
- Developed two-stage XGBoost forecasting model with time-aware validation, improving arrival prediction MAE by 83%

ACTUM Digital

Prague, Czech Republic

Data Analyst and AI Intern

Jun 2025 – Aug 2025

- Built agentic Python pipelines to filter documents, generate metadata, and embed into a Milvus vector database
- Integrated MCP server and prompt workflows for an internal AI assistant, co-presenting the live demo to ACTUM executives
- Automated structured data extraction from 600+ PDFs with Python, consolidating output into a centralized reporting dataset

Million Dollar Baby

Los Angeles, California

Data Analysis Consultant

Apr 2025 – Jun 2025

- Analyzed \$10M+ in advertising spend, surfacing efficiency insights that inform growth and marketing investment strategy
- Presented CAC and RoAS assessments across multiple day windows with Python, guiding decisions on spend allocation
- Built Excel dashboard tracking 20+ marketing KPIs, giving managers a recurring view of acquisition and return trends

TECHNICAL SKILLS

Languages: Python, C++, Java, SQL, R, JavaScript

Libraries/Frameworks: pandas, NumPy, PyTorch, scikit-learn, Matplotlib, TensorFlow, Keras, XGBoost, LangChain, MLflow

Tools & Platforms: AWS, Git, GitHub, GCP, Claude Code, BigQuery, Snowflake, Vertex AI, Tableau, Jupyter Notebooks